BUSINESS & DATA ANALYTICS PORTFOLIO

From a business question to a decision someone can act on

Luciana Jost · TripleTen Certified Data Analyst · Master's in Marketing Management (IPAM Lisboa, in progress)

10 projects · SQL · Python · Statistics · Machine learning · Tableau

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ABOUT

Ten years on the business side, now with the data to back it up

Before data, I spent more than ten years on the business side: marketing for venues and events of up to 3,000 people, brand and positioning consulting, and building my own wellness brand from scratch.

That is where I learned that the hard part is rarely the calculation. It is choosing which question is worth answering and turning the answer into a decision someone can act on.

Every project in this portfolio follows that logic: **a business question at the start, a recommendation at the end**, and the analysis in between documented so anyone can follow it.

Each page shows the question, what I did, what the data showed and what I recommended. Full notebooks, data and presentations are on GitHub.

SKILLS

Data	SQL (PostgreSQL) · Python · pandas · NumPy
Statistics	A/B and A/A/B testing · t-test, Mann-Whitney, z-test, Fisher · Multiple-testing correction
Business analysis	Customer experience · Cohort analysis · Unit economics (LTV, CAC, ROMI) · Conversion funnels · ICE/RICE prioritization · Market analysis
Machine learning	Logistic regression · Random forest · K-Means and hierarchical clustering (scikit-learn)
Visualization	Tableau · Matplotlib · Seaborn · Executive presentations

EDUCATION

TripleTen Data Analyst certification (2026) · Master's in Marketing Management, IPAM Lisboa (in progress) · BA in Fashion Design, UNISINOS · Postgraduate specialization in Event Management, UNISINOS

PROJECTS AT A GLANCE

Ten projects, ten business questions

	PROJECT	BUSINESS QUESTION	MAIN SKILLS	PAGE
01	**Olist: customer experience**	What drives negative reviews in e-commerce, and what weighs most?	Hypothesis testing, text classification, Tableau	4
02	**Y.Afisha: unit economics**	Is marketing spend paying for itself?	Cohorts, LTV, CAC, ROMI	5
03	**Gym churn prediction**	Who is about to cancel, and what keeps members?	Classification, clustering	6
04	**A/B test and hypothesis prioritization**	Which idea to test first, and did the running test win?	ICE/RICE, Mann-Whitney	7
05	**Food delivery funnel and A/A/B test**	Where do users drop out, and is the new font safe?	Funnel, z-test, Bonferroni	8
06	**Book app: SQL analysis**	What can a competitor's database tell a new app?	SQL, CTEs, joins	9
07	**LA cafe market study**	Is there room for a new cafe concept, and in what format?	Market segmentation	10
08	**YouTube trending dashboard**	Where should a video ad team focus?	Tableau, data storytelling	11
09	**Video game market**	Which platforms and genres to bet on next year?	Exploratory analysis, hypothesis testing	12
10	**Zuber rides in Chicago**	Where is demand, and does weather change ride duration?	SQL, t-test	13

PROJECT 01 · CUSTOMER EXPERIENCE · FINAL PROJECT OF THE CERTIFICATION

Olist: what drives negative reviews in e-commerce

Real data from a Brazilian e-commerce marketplace: 95,560 delivered and reviewed orders (Kaggle, 2017 to 2018).

BUSINESS QUESTION

One in eight orders gets a 1 or 2 star review. Which parts of the purchase journey drive it, and which one weighs most?

WHAT I DID

Joined 7 tables, tested four hypotheses from consumer behavior theory (z-test, Mann-Whitney, Fisher, Bonferroni), classified 9,194 negative comments by keyword with hand validation (94% precision) and built a Tableau dashboard.

KEY FINDINGS

- **Breaking the promise hurts more than waiting:** late orders get 24 to 53 pp more negative reviews than on-time orders with a similar wait.
- **70% of late customers review before the product arrives**, because the survey goes out on the estimated date. About 80% of them give a 1 or 2.
- **On time, the problem is the product or a split order:** orders with more than one seller get 39 pp more negative reviews, even when on time.

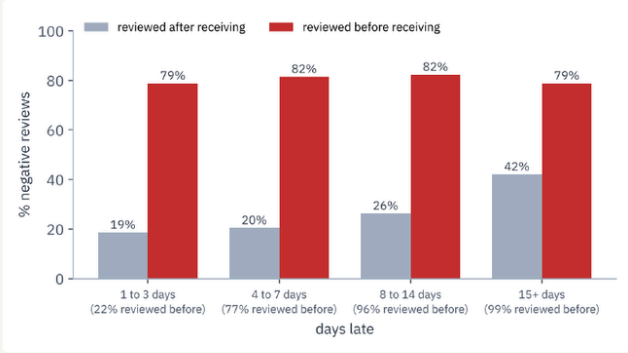
RECOMMENDATION

Send the survey after delivery (with an A/B test of about 385 late orders per group), warn customers before a delay, notify split deliveries and monitor product complaints by seller.

Tools: Python · pandas · SciPy · Matplotlib · Tableau Public

RESULT · H2 SUPPORTED

70% of late customers review before receiving the order



The survey is sent on the **estimated date**. When the order is late, it arrives before the product.

Customers who review without having received it give a 1 or 2 in about **80%** of cases, whatever the size of the delay.

With the product in hand, the rate stays between 19% and 26% for delays of up to 14 days.

Association: the customer chooses when to answer. z-test and Mann-Whitney by delay band, adjusted α of 0.0125.

Live dashboard · Notebook, data and presentation

PROJECT 02 · UNIT ECONOMICS

Y.Afisha: is marketing spend paying for itself?

Marketing efficiency analysis for an entertainment ticketing platform, based on website visits, orders and marketing costs.

BUSINESS QUESTION

Do customers bring in enough value to cover what it costs to acquire them, and where should the budget go?

WHAT I DID

Analyzed product usage, time to first purchase and cohort retention, then calculated LTV, CAC and ROMI by monthly cohort and by marketing source.

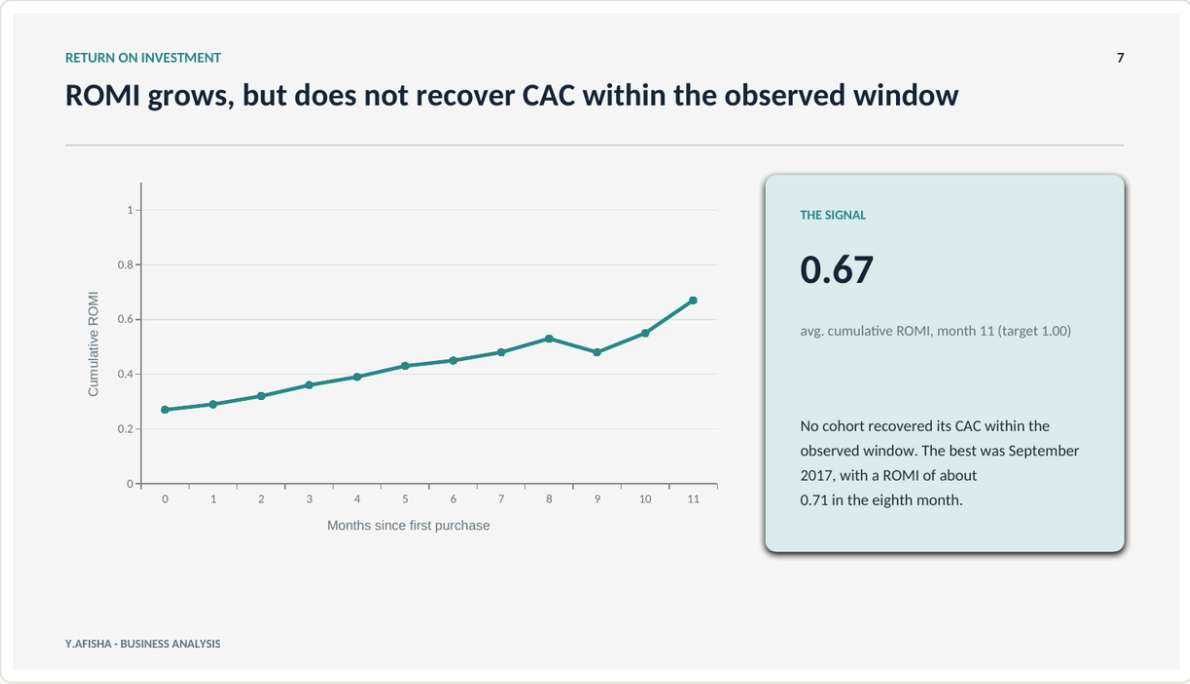
KEY FINDINGS

- **No cohort recovered its acquisition cost** within the observed window. Average cumulative ROMI reached only 0.67 by month 11.
- **Retention collapses in the first month:** only 4.2% to 8.5% of each cohort comes back.
- **Conversion is fast but repeat purchase is rare:** the median customer places a single order.

RECOMMENDATION

Grow value before volume: invest in retention and repeat purchase, and shift acquisition budget toward the sources with the best ratio between customer value and cost.

Tools: Python · pandas · NumPy · Matplotlib · Seaborn



Notebook and presentation on GitHub

PROJECT 03 · MACHINE LEARNING

Gym churn: who is about to cancel, and what keeps members?

Churn prediction and customer segmentation for a gym chain, turned into a retention plan with target groups and concrete actions.

BUSINESS QUESTION

Can we predict which members will cancel next month, and which profiles need attention first?

WHAT I DID

Explored 4,000 member profiles, trained and compared logistic regression and random forest models, and segmented members with hierarchical clustering and K-Means.

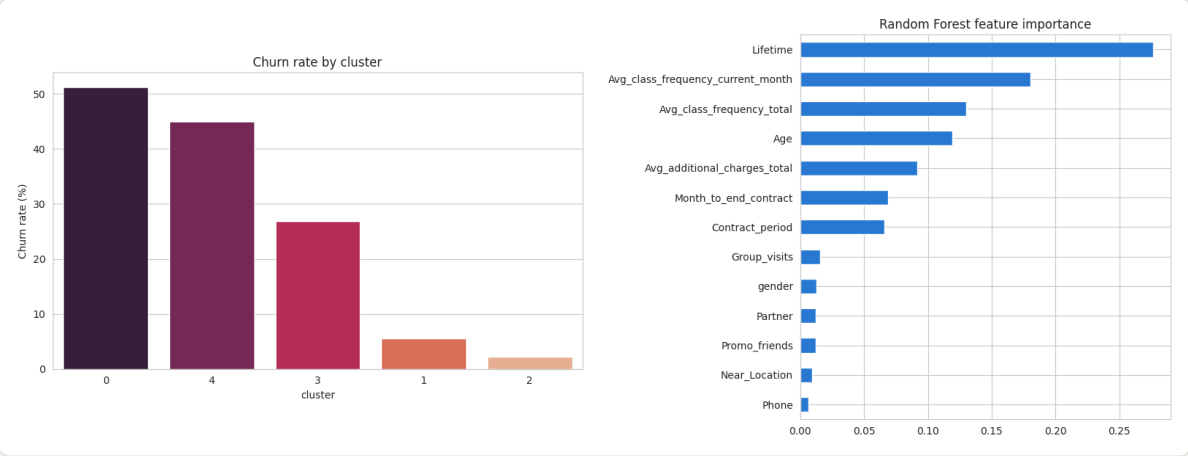
KEY FINDINGS

- **26.5% of members churned.** Logistic regression predicts churn with 92.3% accuracy, 85.4% precision and 82.8% recall, slightly ahead of random forest and easier to explain.
- **Tenure and class attendance** are the strongest signals.
- **New members on short contracts churn at 51.3%**, while members with a partner or referral tie and a long contract churn at 2.2%.

RECOMMENDATION

A structured first 60 days for new short-contract members, incentives to move from monthly to longer contracts, and group classes built into onboarding.

Tools: Python · pandas · scikit-learn · SciPy · Matplotlib · Seaborn



Left: churn rate by customer segment. Right: feature importance in the random forest.

Notebook and data on GitHub

PROJECT 04 · EXPERIMENTATION

Which idea to test first, and did the running A/B test win?

Decision support for an online store: nine marketing hypotheses to prioritize and an A/B test that was still running.

BUSINESS QUESTION

Which hypothesis deserves the next test, and should the current test be stopped?

WHAT I DID

Scored the hypotheses with ICE and RICE, validated the test data, and compared conversion and order value with the Mann-Whitney test, on raw data and without outliers.

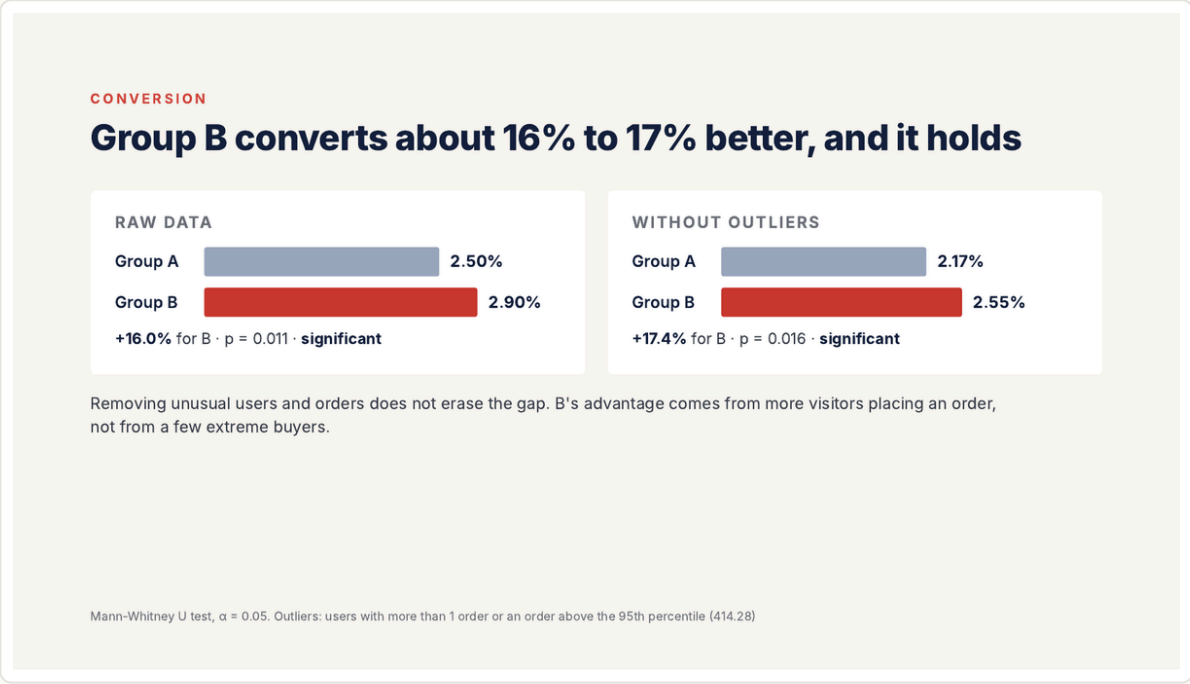
KEY FINDINGS

- **Adding reach changes the ranking:** a subscription form on all main pages moves from third (ICE) to first (RICE).
- **58 users were in both groups** and were removed before any metric was calculated.
- **Group B converts about 16% to 17% better** ($p \approx 0.011$ raw, $p \approx 0.016$ without outliers). The order value gap came from a single huge order and is not significant.

RECOMMENDATION

Stop the test and declare B the winner on conversion, without presenting it as a revenue-per-order gain. Test the subscription form next.

Tools: Python · pandas · NumPy · SciPy · Matplotlib



Notebook and presentation on GitHub

PROJECT 05 · PRODUCT ANALYTICS

Food delivery app: where users drop out, and is the new font safe?

User behavior in a food delivery app: 7,534 users and 240,887 events over one week, plus an A/A/B test of a font change.

BUSINESS QUESTION

Where in the purchase journey do users get lost, and did changing the app's font affect conversion?

WHAT I DID

Built the funnel from main screen to payment using unique users per step, validated the traffic split with an A/A check, and ran two-proportion z-tests across 20 comparisons with Bonferroni correction.

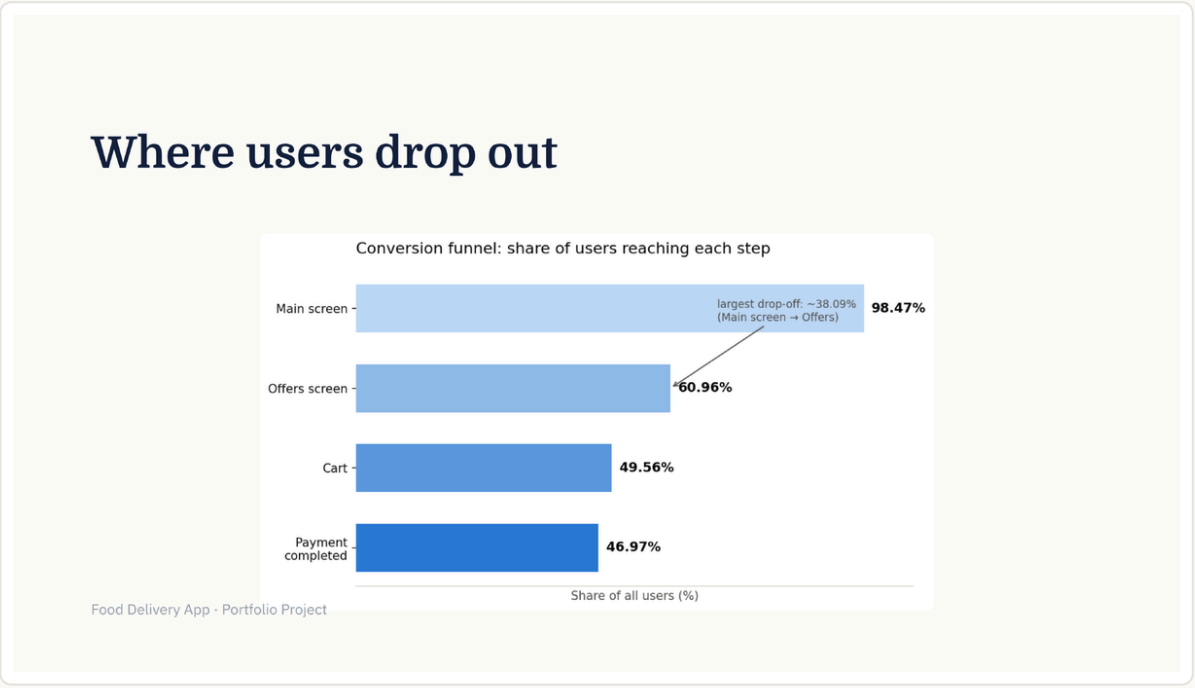
KEY FINDINGS

- **47.7% of users** who see the main screen complete a payment.
- **About 38% never get past the main screen** to the offers screen, by far the steepest drop. Every later step converts above 80%.
- **The font change had no significant effect** on any of the 20 comparisons, even before correcting for multiple tests.

RECOMMENDATION

The font is safe to ship. Focus product work on the step between the main screen and the offers screen.

Tools: Python · pandas · NumPy · Matplotlib · SciPy · statsmodels



Notebook, data and presentation on GitHub

PROJECT 06 · SQL

Book app: what a competitor's database tells a new product

SQL analysis of a competing reading service: 1,000 books, 636 authors, 340 publishers, 6,456 ratings and 2,793 written reviews (PostgreSQL).

BUSINESS QUESTION

What can the competitor's catalog and user behavior tell us about the value proposition of a new reading app?

WHAT I DID

Answered five business questions with SQL: joins, subqueries, CTEs and aggregation with HAVING, taking care not to inflate counts in one-to-many joins.

KEY FINDINGS

- **81.9% of the catalog** was published after 2000.
- **Penguin Books leads** with 42 books over 50 pages; with Penguin Classics, the group reaches 66.
- **Users rate far more than they write:** the six most active users account for only about 5% of all reviews.

RECOMMENDATION (AS HYPOTHESES)

Prioritize recent titles and partnerships with the leading publishing groups, and design features that get casual users to write reviews, not only the most active ones.

Tools: SQL (PostgreSQL) · Python · pandas · SQLAlchemy

```sql
SELECT
    COUNT(*) AS users_over_50_ratings,
    ROUND(AVG(n_reviews), 2) AS avg_reviews
FROM (
    SELECT u.username, COUNT(rv.review_id) AS n_reviews
    FROM (
        SELECT username FROM ratings
        GROUP BY username
        HAVING COUNT(DISTINCT book_id) > 50
    ) AS u
    LEFT JOIN reviews AS rv ON rv.username = u.username
    GROUP BY u.username
) AS per_user;
```

users_over_50_ratings	avg_reviews
6	24.33

Question	Result
Books released after Jan 1, 2000	819 (81.9%)
Reviews per book	0 to 7, average 2.79
Top publisher, books over 50 pages	Penguin Books, 42
Best-rated author (books with 50+ ratings)	J.K. Rowling/Mary GrandPré, 4.28

Query from section 3.5, shown with English aliases for readability. The notebook keeps the original run.

Notebook on GitHub

PROJECT 07 · MARKET ANALYSIS

LA cafe market: is there room for a new concept, and in what format?

Market viability study for a robot-service cafe in Los Angeles, across 9,650 establishments.

BUSINESS QUESTION

Is the market open to a new cafe concept, and what should the first unit look like to attract investors?

WHAT I DID

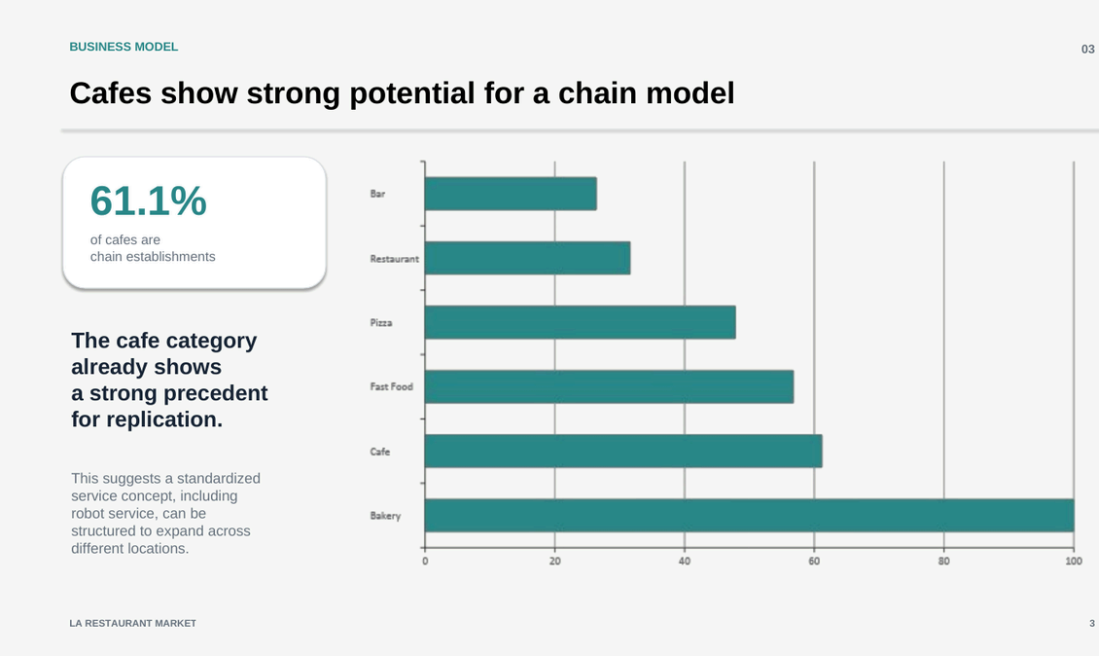
Segmented the market by establishment type, compared chains and independents, benchmarked seat capacity and mapped street-level density.

KEY FINDINGS

- **Restaurants dominate** (75.2% of the market); cafes are only 4.5%, a small category where a differentiated concept can stand out.
- **61.1% of cafes belong to chains**, the second highest share after bakeries, which shows that the format can be replicated.
- **Chains favor compact units** of about 25 to 30 seats rather than large flagships.

RECOMMENDATION

Start small, design to scale: one efficient unit of 25 to 30 seats, with robot service as the experience differentiator, built as the model for a chain.



Notebook and presentation on GitHub

Tools: Python · pandas · Seaborn · Matplotlib

PROJECT 08 · DASHBOARD

YouTube trending: where should a video ad team focus?

Interactive Tableau dashboard built on 339,990 trending video records from 5 countries, November 2017 to June 2018.

BUSINESS QUESTION

Which categories and countries dominate trending videos, and how should an ad team plan by market?

WHAT I DID

Designed a dashboard with four linked views and filters for date and country, and wrote the takeaways for the ad planning team.

KEY FINDINGS

- **Entertainment is 27.9%** of all trending videos, more than double the next category.
- **In the US, Music and Howto & Style weigh about twice as much** as in other markets.
- **The February 2018 jump in volume** came from Japan entering the data, not from real growth.

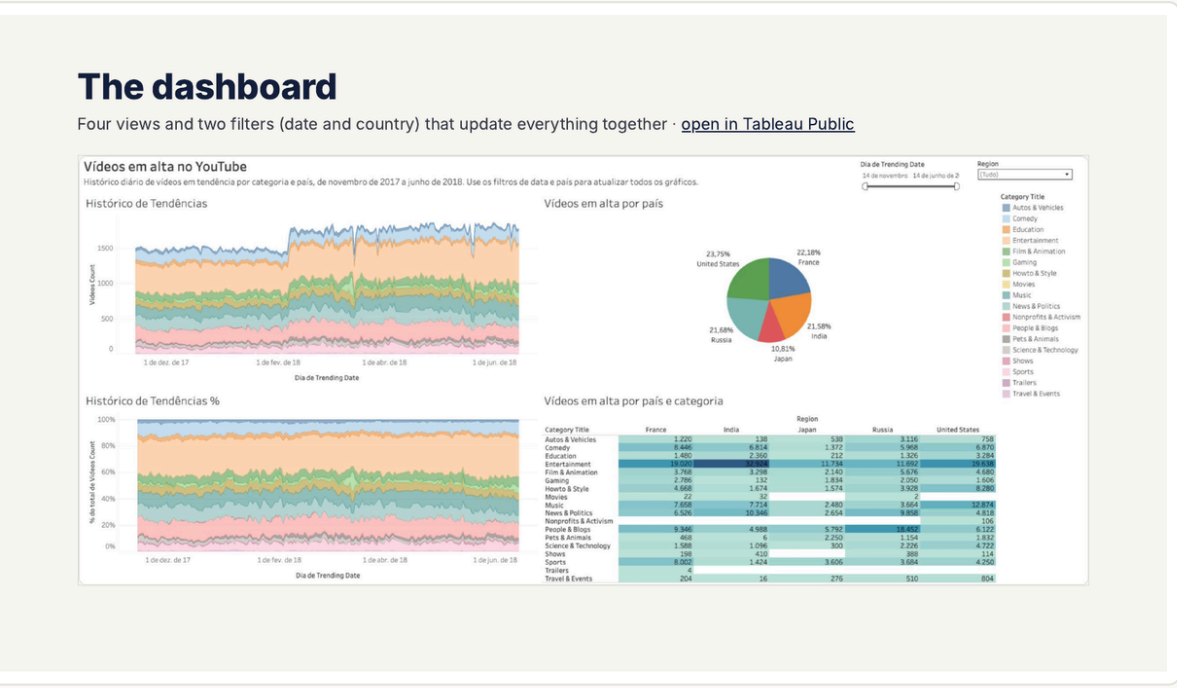
RECOMMENDATION

Use Entertainment as the safe bet everywhere, prioritize Music, Howto & Style and Science & Technology in the US, and filter by country when comparing periods.

Tools: Tableau Public

The dashboard

Four views and two filters (date and country) that update everything together · open in Tableau Public



Live dashboard · **Presentation and data on GitHub**

PROJECT 09 · EXPLORATORY ANALYSIS

Video game market: which platforms and genres to bet on?

Analysis of 16,715 games across platforms, genres and regions to guide an online store's 2017 campaign planning.

BUSINESS QUESTION

Which platforms and genres are most likely to sell next year, and does one strategy work for every region?

WHAT I DID

Analyzed platform life cycles, sales by genre and region, and the link between reviews and sales, and tested two hypotheses about user ratings.

KEY FINDINGS

- **PS4 leads current-generation sales**, with Xbox One as the second bet.
- **Critic scores show a moderate link to sales** ($r \approx 0.41$); user scores show none.
- **Japan behaves as a separate market**, led by handheld consoles and role-playing games.

RECOMMENDATION

Bet on PS4 and Xbox One, lead with Shooter and Action, invest in critical reception, and run a distinct strategy for Japan.

Tools: Python · pandas · Matplotlib · Seaborn · SciPy



Notebook and presentation on GitHub

PROJECT 10 · SQL AND HYPOTHESIS TESTING

Zuber: where is demand in Chicago, and does weather matter?

Exploratory analysis and hypothesis testing for a new ride-sharing company preparing to launch in Chicago.

BUSINESS QUESTION

How concentrated is the market, where do rides end, and does weather change trip duration to the airport?

WHAT I DID

Extracted data with SQL, analyzed 64 taxi companies and 94 drop-off neighborhoods, and ran a two-sample t-test on Loop to O'Hare rides on Saturdays.

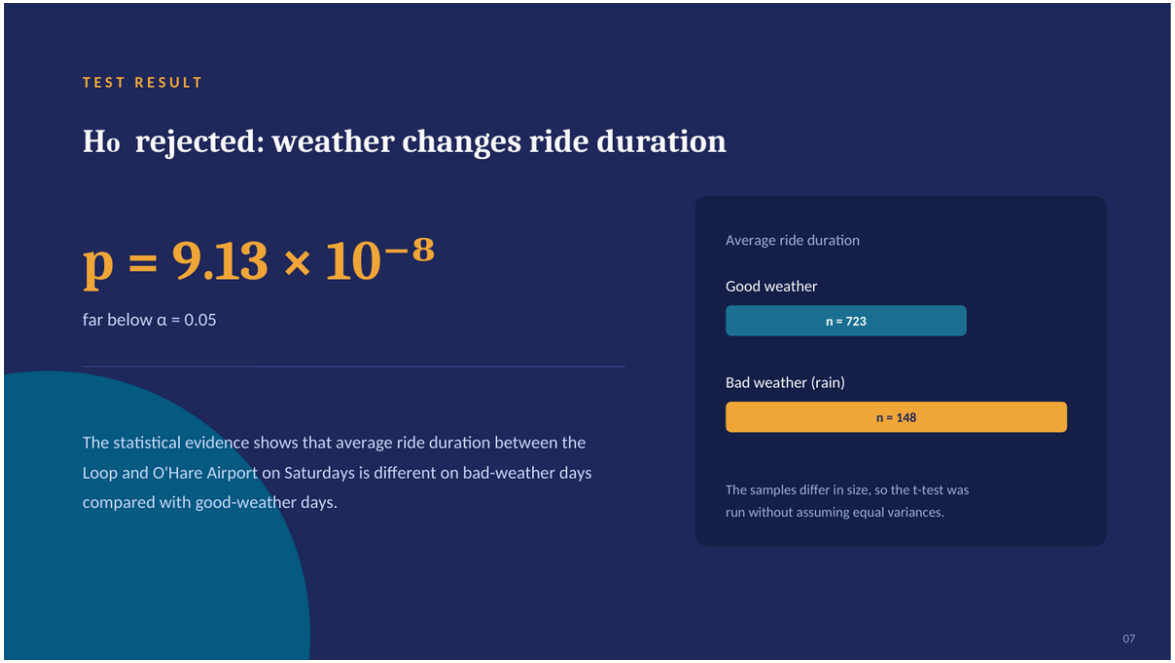
KEY FINDINGS

- **Flash Cab leads** with 19,558 rides; the top three companies account for about 30% of all rides.
- **The Loop is the top destination**, with 10,727 rides on average.
- **Rides to O'Hare take significantly longer on rainy Saturdays** ($p \approx 9.13 \times 10^{-8}$).

RECOMMENDATION

Allocate the fleet by neighborhood demand and treat weather as a variable in pricing and arrival time estimates for airport trips.

Tools: SQL · Python · pandas · Matplotlib · SciPy



Notebook and presentation on GitHub

LET'S TALK

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LinkedIn	linkedin.com/in/lujost
GitHub portfolio	github.com/lucianajost/data_analytics_portfolio
Olist dashboard	Tableau Public